

CLASS 11TH PHYSICS | CRUCIAL BOARD QUESTIONS (CYQs) EXAM QUESTION BANK

UNIT 1: UNITS & DIMENSIONS

- Q1.** Define physical quantity.
- Q2.** Define unit and show that the numerical value of a physical quantity is inversely proportional to the unit of the physical quantity.
- Q3.** What are fundamental and derived units?
- Q4.** Why are mass, length, and time chosen as fundamental quantities?
- Q5.** Name various systems of units. Write fundamental or base units and supplementary units in SI.
- Q6.** Write the advantages of SI units.
- Q7.** Define (i) dimensions, (ii) dimensional formula, and (iii) dimensional equation of a physical quantity. Also define the principle of homogeneity.
- Q8.** Write the dimensional formula of Planck's constant, torque, surface tension, angular momentum, velocity gradient, modulus of rigidity, etc.
- Q9.** What are the uses of dimensional analysis?
- Q10.** Find dimensions of constants a and b in the equation:
 $(P + a/V^2)(V - b) = RT$
- Q11.** Write the dimensions of a and b in the relation:
 $E = (b - x^2) / (a * t)$
where E is energy, x is distance, and t is time.
- Q12.** A force is given by $F = at + bt^2$, where t is time. The dimensions of a and b are respectively:
- Q13.** Convert 10 newton into dyne using dimensional analysis.
- Q14.** Convert 5 joule into erg using dimensional analysis.
- Q15.** Test the correctness of the equation: $S = ut + (1/2)at^2$.
- Q16.** Check the accuracy of the relation $V = (P * r^4) / (8 * \eta * l)$, where V is the volume per unit time, P is the pressure, r is the radius, η is the coefficient of viscosity, and l is the length.
- Q17.** Suppose that the oscillations of a simple pendulum depend on:
(I) the length of the string (L)
(II) acceleration due to gravity (g)
Dimensionally, show which of the above factors have an influence upon the period and in what way.

- Q18.** A body of mass m moves in a circular path of radius r with velocity v . Find the expression for the centripetal force F acting on the particle using the method of dimensions.
- Q19.** The kinetic energy possessed by a body depends upon its (i) mass (m) (ii) speed (v). Find the expression for the kinetic energy of the body using the method of dimensions.
- Q20.** The frequency (ν) of an oscillating liquid drop may depend upon radius (r) of the drop, density (ρ) of liquid, and the surface tension (S). Derive the formula dimensionally.
- Q21.** A solid sphere moves through a liquid. The liquid opposes the motion of the sphere with a force F which depends on the radius r , speed v of the sphere, and coefficient of viscosity η of the liquid. Derive the expression for force.
- Q22.** What are the limitations of dimensional analysis?
- Q23.** State the number of significant figures in the following:
28, 0.005 m², 2.63 $\times 10^{28}$ kg, 0.2560 cm³, 6.320 N, 5.003 J, 0.0006032, 200.05, 12500, 122, 0.08240
- Q24.** What is the difference between 5 and 5.00?
- Q25.** If $x = a + bt + ct^2$ where x is in metres and t in seconds, what are the units of b and c ?

UNIT 2: VECTORS

- Q26.** Define scalar and vector quantities. Give examples of each.
- Q27.** Define unit vector.
- Q28.** Which of the following are meaningful?
(1) Addition of any two scalars.
(2) Addition of a scalar to a vector of the same dimension.
(3) Multiplication of any vector by any scalar.
(4) Multiplication of any two scalars.
(5) Addition of two vectors.
- Q29.** Does the nature of a vector change when it is multiplied by a scalar?
- Q30.** If a physical quantity has both magnitude and direction, is it necessarily a vector? Is rotation always a vector quantity?
- Q31.** State the Triangle law of vector addition. Give its analytical treatment to find the magnitude and direction of a resultant vector.
- Q32.** State the Parallelogram law of vector addition. Find analytically the magnitude and direction of resultant vector when (i) two vectors are parallel to each other, (ii) two vectors are perpendicular to each other.
- Q33.** Explain the scalar product of two vectors. Mention any two of its properties.
- Q34.** Explain the cross product of two vectors. Give its examples and properties.

UNIT 3: KINEMATICS

Q35. Derive:

(i) $v = u + at$

(ii) $s = ut + \frac{1}{2}at^2$

(iii) $v^2 - u^2 = 2as$

Q36. Define displacement of a particle. What is the difference between distance and displacement? What are the characteristics of displacement?

Q37. Is the magnitude of displacement always equal to distance? Under what conditions will it be equal?

Q38. Is it possible that the displacement of a moving particle is zero but not the distance? Explain.

Q39. What does the slope of a position-time graph represent?

Q40. Distinguish between speed and velocity.

Q41. What is the difference between speed and velocity? Show that the slope of a displacement-time graph is equal to the velocity of uniform motion.

Q42. Can a body be said to be at rest as well as in motion at the same time? Explain.

Q43. If the displacement-time graph of a particle is parallel to the time axis, what will be the velocity of the particle?

Q44. Show that the displacement of a particle in a given interval of time is equal to the area under the velocity-time graph of uniform motion.

Q45. Define uniformly accelerated motion.

Q46. Is the magnitude of average velocity always equal to average speed? Under what condition will average velocity and average speed be the same?

Q47. Draw a model graph between:

(I) Displacement and time

(II) Velocity and time for the motion of a particle moving with uniform velocity.

UNIT 4: MOTION IN A PLANE

Q48. Define projectile motion. Derive an expression for maximum height, time of flight, horizontal range, and the equation of trajectory of a projectile projected at an angle with the horizontal.

Q49. At what point in its trajectory does a projectile have minimum speed?

Q50. When a projectile is projected at an angle with the horizontal, which component of the velocity remains constant throughout the trajectory?

Q51. Define angular displacement, angular speed, and angular acceleration in circular motion. How are linear velocity and linear acceleration related to angular speed and angular acceleration respectively?

UNIT 5: LAWS OF MOTION & FRICTION

Q52. What is inertia? Define and explain the three different types of inertia.

- Q53.** Why do we beat a carpet with a stick to remove dust particles?
- Q54.** Why does a passenger fall backward when a bus suddenly starts moving from rest?
- Q55.** A man jumping out of a moving bus falls with his head in the forward direction. Explain.
- Q56.** If we jerk a piece of paper placed under a book quickly, the paper moves out without the movement of the book. Why?
- Q57.** What is force? What are the absolute and gravitational units of force in CGS and SI systems? How are these different units related to each other?
- Q58.** Explain Newton's first law of motion. Why do we call it the law of inertia?
- Q59.** State Newton's second law of motion. How does it help to measure force? Also show that Newton's second law of motion is the real law of motion.
- Q60.** State Newton's third law. Action and reaction are equal and opposite; why can they not cancel each other?
- Q61.** How does Newton's third law of motion help us in walking?
- Q62.** When a man jumps out of a boat, the boat is pushed backward. Why?
- Q63.** Define Impulse and derive the Impulse-Momentum theorem.
- Q64.** Why does a cricket player lower his hands while catching a cricket ball?
- Q65.** State and prove the law of conservation of linear momentum and illustrate it with an example.
- Q66.** What is the principle of rocket propulsion?
- Q67.** Why does a heavy gun not recoil as strongly as a light gun firing the same bullet?
- Q68.** Explain various types of friction. State the laws of limiting friction. Give some methods for reducing friction.
- Q69.** Sand is thrown on roads covered with snow. Explain why.
- Q70.** Define friction. Explain static friction, limiting friction, and kinetic friction.
- Q71.** Define the angle of friction and angle of repose, and show that both are numerically equal.
- Q72.** What is the need for banking a road? Obtain an expression for the maximum speed with which a vehicle can safely negotiate a curved road banked at an angle.
- Q73.** Why does a cyclist bend inwards while negotiating a curve? Explain with a diagram.

UNIT 6: WORK, POWER AND ENERGY

- Q74.** State and explain the work-energy theorem.
- Q75.** Derive an equation for potential energy stored in a stretched spring.
- Q76.** What are conservative and non-conservative forces? Explain with examples.

- Q77.** Write down the condition for the conservation of mechanical energy of a system.
- Q78.** Derive the relationship between kinetic energy and linear momentum: $KE = p^2 / (2m)$.
- Q79.** Define work, power, and energy. Give their SI units.
- Q80.** What is power? Prove that power is the dot product of force and velocity.
- Q81.** What is meant by mechanical energy? What type of energy is stored in the spring of a watch?
- Q82.** When a body moves with constant velocity, no work is done. Explain.
- Q83.** Define potential energy and kinetic energy.
- Q84.** Define centripetal force.
- Q85.** Why does a coolie do no work when he moves on a level road carrying a box on his head?
- Q86.** The work done in extending a spring by x is W . Find the work done if the spring is further extended by x .
- Q87.** What should be the angle between force and displacement vectors for the force to do maximum work?
- Q88.** What do you understand by positive and negative work?
- Q89.** How will the momentum of a body change if its kinetic energy is doubled?
- Q90.** What does the area under a force-displacement graph represent?
- Q91.** What is the work done by centripetal force when a body completes one rotation around a circle?
- Q92.** A light body and a heavy body have equal kinetic energies. Which one has larger momentum?
- Q93.** How many watts are in one horsepower?
- Q94.** How many joules are in 1 eV?

UNIT 7: CENTRE OF MASS, MOMENTUM AND COLLISION

- Q95.** Define elastic collision, inelastic collision, and coefficient of restitution. How does the coefficient of restitution define elastic and inelastic collisions?
- Q96.** State & explain the law of conservation of linear momentum. Explain the recoil of a gun, explosion of a bomb, etc.
- Q97.** Define centre of mass. Calculate the centre of mass of a two-particle system.
- Q98.** Prove that in an elastic collision in one dimension, the relative velocity of approach before impact is equal to the relative velocity of separation after impact.
- Q99.** Why does a cricketer lower his hands while catching a cricket ball?
- Q100.** A person falling on a cemented floor gets injured more than when falling on a heap of sand. Why?

- Q101.** When a ball falls on earth, the earth also moves up to meet it, but the motion of the earth is not noticeable. Why?
- Q102.** Define impulse and derive the equation for it. What does the area under a force-time curve represent?
- Q103.** Briefly analyse elastic collision in two dimensions.
- Q104.** Why is the recoil of a gun when fired based on the law of conservation of linear momentum?
- Q105.** What remains conserved in an inelastic collision?
- Q106.** What remains conserved in an elastic collision?
- Q107.** If one mass is heavier than the other, toward which mass will the centre of mass shift?
- Q108.** For which body is the centre of mass outside the body: A) Ring B) Disc?
- Q109.** What does the area under a force-time curve represent?
- Q110.** An astronaut in open space is away from his spaceship. How can he return to his ship? Why does a heavy gun not recoil as strongly as a light gun firing the same bullet?
- Q111.** A body of mass m moving with velocity V collides elastically with another body B of the same mass at rest. What will be their velocities after collision?
- Q112.** When is the exchange of energy maximum in an elastic collision?
- Q113.** Can the geometrical centre and centre of mass of a body coincide? If yes, give examples.
- Q114.** Define coefficient of restitution & write its value for perfectly elastic and inelastic collisions.
- Q115.** If two bodies stick together after collision, will the collision be elastic or inelastic?

UNIT 8: ROTATIONAL MOTION AND CIRCULAR MOTION

- Q116.** Define angular momentum. Derive the relationship between angular momentum & torque.
- Q117.** Why are curved roads banked? Obtain an expression for the angle of banking.
- Q118.** Explain the conservation of angular momentum.
- Q119.** Derive a relation for torque acting on a particle in a plane.
- Q120.** Define centripetal force. Derive an expression for it.
- Q121.** A car of mass 90 kg is travelling around a circular path of radius 300 m with speed 72 km/h. Calculate its angular momentum.
- Q122.** A stone weighing 3 kg and tied to a string is rotated in a horizontal circle of radius 120 cm with a velocity of 500 m/s. Find the total force which tries to break the string.
- Q123.** Explain why it is difficult to open a door by pushing or pulling it near the hinge.

- Q124.** What should be the minimum velocity at the lowest point to complete a circle in a vertical plane?
- Q125.** What is the work done by centripetal force? Does the moment of inertia of a body change with the speed of rotation?
- Q126.** What is the physical significance of moment of inertia? On what factors does moment of inertia depend?
- Q127.** What is the need for banking of tracks?
- Q128.** State two conditions for the equilibrium of a rigid body.
- Q129.** Divers leave the springboard with arms and legs outstretched and then pull them inwards to execute somersaults. Explain.
- Q130.** How does a ballet dancer or ice skater increase or decrease their rate of spinning?
- Q131.** If all the polar ice caps of the earth melt, how will it affect the duration of the day?
- Q132.** If the earth contracts to half its radius, what would be the duration of the day?
- Q133.** A stone tied at the end of a string is whirled in a horizontal circle. When the string breaks, the stone flies off tangentially. Why?
- Q134.** Define radius of gyration. Write an expression for it.
- Q135.** What is a rigid body?
- Q136.** In which type of circular motion will the object possess both centripetal and transverse (tangential) acceleration?
- Q137.** What are centripetal and tangential acceleration?
- Q138.** A body is moving along a circle such that its speed is constant. Should there be a force acting on the body?
- Q139.** If the net external torque acting on a system is zero, what can we say about its angular momentum?

UNIT 9: GRAVITATION

- Q140.** State and explain Kepler's laws of planetary motion.
- Q141.** Derive an expression for the orbital velocity of a satellite.
- Q142.** Derive an equation for gravitational potential.
- Q143.** Derive an equation for gravitational potential energy.
- Q144.** Derive an expression for acceleration due to gravity (g) on the surface of the earth.
- Q145.** Derive how acceleration due to gravity varies with height and depth from the surface of the earth.
- Q146.** Define escape velocity and derive an equation for it.
- Q147.** Define orbital velocity. Derive an expression for it.

- Q148.** The escape speed on earth is 11.2 km/s. Find the escape speed on another planet of mass 100 times and radius 10 times that of the earth.
- Q149.** Why are space rockets generally launched from west to east?
- Q150.** Deduce Newton's law of gravitation from Kepler's laws.
- Q151.** A man can jump on the moon's surface 6 times as high as on earth. Why? Explain why a 5 kg mass weighs more at the poles than at the equator.
- Q152.** When an apple falls towards the earth, the earth also moves up to meet the apple. Is it true? Explain.
- Q153.** The earth is continuously pulling the moon towards its centre. Why does the moon not fall onto the earth?
- Q154.** At midnight, the sun pulls us in the same direction as the earth, but at midday, the sun pulls in the direction opposite to the earth. Would we weigh more at midnight? Explain.
- Q155.** What will happen to the time period of a simple pendulum if it is taken from the surface of the earth to the moon?
- Q156.** What is the reason for weightlessness in an orbiting satellite?
- Q157.** Why is Newton's law of gravitation called a universal law?
- Q158.** Define universal gravitational constant (G).
- Q159.** What is the difference between inertial mass and gravitational mass?
- Q160.** Distinguish between gravity and gravitation.
- Q161.** What is the weight of a body at the centre of the earth?

UNIT 10: MECHANICS OF SOLIDS & FLUIDS

- Q162.** State Hooke's law and define Young's modulus of elasticity.
- Q163.** State Stokes' law and derive the expression for viscous force acting on a spherical body falling through a viscous fluid using dimensional analysis.
- Q164.** State the law of floatation.
- Q165.** Define capillarity. Find an expression for the ascent of liquid in a capillary tube (ascent formula).
- Q166.** State Bernoulli's theorem and mention its applications.
- Q167.** Define terminal velocity and derive a relation for it.
- Q168.** Define surface tension and derive its relation with surface energy.
- Q169.** State Pascal's law and discuss its applications.
- Q170.** State Bernoulli's theorem in mathematical form.
- Q171.** Give the factors on which surface tension depends.

- Q172.** Define streamline and turbulent flow.
- Q173.** Why does oil spread over the surface of water whereas water does not spread over the surface of oil?
- Q174.** Derive an expression for excess pressure: (i) inside a liquid drop, (ii) inside a soap bubble.
- Q175.** Define Bulk modulus, Modulus of rigidity, and Young's modulus.
- Q176.** Define elasticity.
- Q177.** Define elastic limit.
- Q178.** Define Poisson's ratio. Give its theoretical and practical limits.
- Q179.** Define elastic fatigue.
- Q180.** Why is steel more elastic than rubber?
- Q181.** Why is it easier to cut with a sharp knife than a blunt knife?
- Q182.** Why is it easier to swim in sea water than in river water?
- Q183.** Why does a ship made of iron float in water although an iron needle sinks?
- Q184.** During a windstorm, light roofs are often blown off. Why?
- Q185.** A person standing close to a railway track may get pulled toward a fast-moving train. Explain.
- Q186.** What are the characteristics of an ideal fluid?
- Q187.** Define the following terms: (a) Cohesion, (b) Adhesion, (c) Molecular range, (d) Sphere of influence.
- Q188.** What happens to surface tension when impurities are added to a liquid?
- Q189.** Explain why small liquid drops are spherical in shape while large drops are flattened.
- Q190.** A sewing needle can float on clean water, but it sinks when detergent is added. Comment.
- Q191.** Why does hot soup taste better than cold soup?
- Q192.** Two soap bubbles of unequal sizes are blown at the ends of a connecting tube. Which one will grow at the expense of the other?
- Q193.** How does surface tension vary with temperature?
- Q194.** How are small insects able to walk on the surface of water?
- Q195.** Why should antiseptics have low surface tension?
- Q196.** When two drops of mercury are brought into contact, they merge into one drop. Why?
- Q197.** Why does a hot soap solution clean clothes better than a cold solution?
- Q198.** Why does a new earthen pot keep water cooler than an older one?

Q199. An ice cube floats in a glass of water filled to the brim. What happens to the water level when the ice melts?

Q200. Why are the upper surfaces of aeroplane wings curved while the lower surfaces are relatively flat?

UNIT 11: KTG, THERMODYNAMICS & THERMAL PROPERTIES

Q201. Derive an expression for the pressure exerted by an ideal gas in terms of root mean square velocity.

Q202. Derive expressions for work done in isobaric, isothermal, and adiabatic processes.

Q203. Write down four postulates of the kinetic theory of gases.

Q204. Explain Wien's displacement law.

Q205. Define isothermal and adiabatic processes.

Q206. State the zeroth law of thermodynamics. How does this law define temperature?

Q207. Define degrees of freedom. Calculate the degrees of freedom for monoatomic, diatomic, and triatomic gases.

Q208. Discuss the kinetic interpretation of temperature. What is absolute zero temperature?

Q209. State the law of equipartition of energy.

Q210. What are reversible and irreversible processes?

Q211. What are the various modes of heat transfer? Discuss them in detail.

Q212. State the second law of thermodynamics.

Q213. State the first law of thermodynamics in mathematical form.

Q214. Prove that $C_p - C_v = R$ (Mayer's relation).

Q215. Discuss the limitations of the first law of thermodynamics.

Q216. Can the temperature of a gas be increased while keeping its pressure and volume constant?

Q217. Define the principle of calorimetry.

Q218. Distinguish between isothermal and adiabatic changes.

Q219. The speeds of 5 molecules of a gas are 2, 3, 4, 5, and 6 m/s. What is their root mean square velocity?

Q220. Define coefficient of linear expansion, coefficient of superficial expansion, and coefficient of cubical expansion.

Q221. What is the relation between degrees of freedom and specific heat ratio (γ)?

Q222. In which process is work done maximum: (a) isothermal, (b) adiabatic, (c) isobaric, or (d) isochoric?

Q223. What is the anomalous expansion of water?

Q224. What is a perfect black body?

Q225. State Stefan-Boltzmann law.

UNIT 12: OSCILLATIONS AND WAVES

Q226. Distinguish between periodic and oscillatory motion. Give two examples of each.

Q227. Define simple harmonic motion (SHM). What are its characteristics? Derive expressions for:

- (a) Displacement
- (b) Velocity
- (c) Acceleration of a particle in SHM.

Q228. Show that the total energy of a particle executing SHM is directly proportional to the square of amplitude and frequency.

Q229. Discuss graphically the variation of kinetic energy, potential energy, and total energy with displacement for a particle executing SHM.

Q230. What is a simple pendulum? Show that the motion of a simple pendulum is SHM and deduce an expression for its time period. Also define a seconds pendulum.

Q231. What is the total distance travelled by a body executing SHM in a time equal to its time period, if its amplitude is A ?

Q232. Will a pendulum clock gain or lose time when taken to the top of a mountain?

Q233. Discuss the formation of standing waves on a stretched string fixed at both ends and describe its different modes of vibration.

Q234. Under what condition is the motion of a simple pendulum simple harmonic?

Q235. What is wave motion? Describe the characteristics of wave motion.

Q236. Define periodic motion.

Q237. Define oscillatory motion.

Q238. Are all periodic motions oscillatory in nature? Explain.

Q239. What is the basic condition for a motion to be simple harmonic?

Q240. Distinguish between transverse and longitudinal waves. Give two examples of each.

Q241. What do you mean by wavelength, frequency, and amplitude of a wave?

Q242. Discuss Newton's formula for the velocity of longitudinal waves in a medium and Laplace's correction.

Q243. What is a progressive wave? Derive the equation of a plane progressive harmonic wave.

Q244. What are the characteristics of stationary waves? Define nodes and antinodes.

Q245. Distinguish between progressive waves and stationary waves.

Q246. What are beats? Define beat frequency and beat period.

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